

CLINTON, IA, USA

WMO: 725473

Lat: 41.833N

Lon: 90.333W

Elev: 213

StdP: 98.79

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-18.8	-27.3	0.3	-17.6	-23.6	0.5	-16.4	14.1	-4.3	12.6	-7.6	3.5	280	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	32.6	24.0	31.3	23.4	29.7	22.6	25.8	30.8	24.8	29.5	23.8	28.0	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.5	28.9	23.0	18.2	27.5	22.4	17.5	26.9	81.1	31.0	76.2	29.4	72.3	28.3	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	11.1	9.5	DB	-25.4	34.8	4.1	1.5	-28.3	35.9	-30.8	36.8	-33.1	37.6	-36.1	38.7	
			WB	-26.1	27.4	3.8	1.1	-28.8	28.3	-31.1	28.9	-33.2	29.6	-36.0	30.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-6.2	-3.9	2.8	9.5	16.1	21.3	23.1	21.7	18.0	11.0	3.9	-2.8
	DBStd	11.46	7.01	6.84	6.49	5.25	4.69	3.69	3.09	2.90	4.31	5.33	5.76	6.42
	HDD10.0	1852	501	390	238	70	6	0	0	0	2	52	195	398
	HDD18.3	3637	759	623	482	269	102	14	2	6	59	235	432	654
	CDD10.0	1708	0	1	16	55	194	338	405	362	241	83	13	2
	CDD18.3	451	0	0	1	4	32	102	149	108	47	7	0	0
	CDH23.3	4117	0	0	8	58	303	965	1396	874	444	70	1	0
	CDH26.7	1333	0	0	1	11	78	341	504	256	129	14	0	0
Wind	WSAvg	4.5	4.9	5.0	5.3	5.8	5.0	4.1	3.2	3.0	3.5	4.4	4.9	4.7
Precipitation	PrecAvg	881	38	33	60	83	100	109	94	109	86	67	56	46
	PrecMax	1281	119	85	167	181	314	372	222	350	284	165	138	106
	PrecMin	498	2	4	11	19	19	23	4	25	3	0	8	9
	PrecStd	175	24	23	37	40	59	67	48	74	59	45	32	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	15.9	24.0	28.0	32.0	33.7	35.3	33.0	32.5	28.7	22.0	15.9
		MCWB	8.9	10.9	16.8	17.3	21.9	23.6	26.4	24.7	23.1	22.0	15.1	13.8
	2%	DB	7.4	11.1	19.1	24.8	28.7	32.3	32.7	31.2	30.0	25.1	17.8	11.9
		MCWB	5.1	8.2	13.4	15.8	19.4	23.0	24.8	24.5	21.9	18.3	12.6	9.7
	5%	DB	4.1	7.5	16.1	22.0	27.1	30.7	31.3	29.1	27.8	22.5	15.2	7.9
		MCWB	2.5	5.1	11.7	14.2	18.6	22.2	24.2	23.4	20.4	16.5	11.1	6.0
	10%	DB	2.5	4.1	12.5	18.8	24.2	28.4	29.1	27.7	26.2	19.8	12.7	5.3
		MCWB	1.2	2.2	8.6	12.3	17.2	21.3	23.1	22.2	19.3	15.3	9.2	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.9	17.5	19.2	23.0	25.8	27.8	26.8	25.0	22.2	16.2	14.2
		MCDB	10.9	13.9	23.9	25.8	29.4	31.6	32.7	31.2	29.7	27.4	20.0	16.1
	2%	WB	5.2	8.0	14.8	16.9	21.4	24.4	26.4	25.4	23.2	19.6	13.7	9.5
		MCDB	7.4	10.3	18.4	22.9	26.7	29.8	31.5	29.8	27.7	23.4	16.9	11.2
	5%	WB	2.9	5.1	11.9	15.2	19.9	23.2	25.2	24.2	21.8	17.7	11.8	6.5
		MCDB	4.3	7.4	15.8	20.7	24.9	28.4	30.2	28.1	26.3	21.0	14.8	8.2
	10%	WB	1.2	2.7	8.7	13.1	18.6	22.0	24.1	23.2	20.5	15.5	9.6	3.6
		MCDB	2.4	4.3	12.4	18.3	23.2	27.2	28.2	26.8	24.7	19.2	12.7	5.3
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.2	10.3	12.0	11.8	11.3	11.2	11.0	12.7	11.6	10.4	8.4
		MCDBR	10.2	11.7	14.3	16.7	13.9	12.9	12.3	12.1	14.3	14.3	13.5	11.0
		MCWBR	8.4	9.0	9.5	9.9	7.6	6.2	6.6	6.6	7.4	8.6	9.4	9.0
	5% WB	MCDBR	9.9	11.4	13.5	14.2	12.0	11.4	11.1	10.6	11.8	12.0	12.3	10.4
		MCWBR	8.3	9.0	9.5	9.4	7.5	6.3	6.7	6.5	6.7	8.4	9.5	9.0
Clear-Sky Solar Irradiance	taub	0.290	0.301	0.341	0.387	0.416	0.424	0.442	0.421	0.392	0.343	0.301	0.287	
	taud	2.404	2.384	2.371	2.255	2.239	2.263	2.200	2.259	2.312	2.451	2.502	2.432	
	Ebn at Noon	862	908	905	881	861	853	832	839	838	848	846	834	
	Edh at Noon	79	94	108	131	137	134	141	129	114	88	72	71	
All-Sky Solar Radiation	RadAvg	1.75	2.74	3.73	4.65	5.46	6.13	6.09	5.37	4.52	2.97	1.96	1.43	
	RadStd	0.21	0.30	0.33	0.40	0.39	0.51	0.39	0.35	0.37	0.35	0.16	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page